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A Model calibration and estimation

In Table A.1 we report the countries considered in our analysis.

The output of each country is provided by the gross domestic product per capita, measured by purchasing power parity (GDP PPP) as reported by the International Monetary Fund (IMF)[3]. The data analysis reveals significant disparities in output levels across countries. To enhance comparability across the countries of the three macro-regions, we normalize both output and debt relative to the regional sample average for the period 2013-2023, expressed in dollars (SSA: $5,460$\$, LATAM: $16,367$\$, SA-APAC: $8,946$\$).

A.1 Pollution process

As a measure of pollution stock, we consider the Climate-driven INFORM Risk indicator, computed by the IMF for a panel of 188 countries on an annual basis since 2013[4]. The indicator has three dimensions (climate-driven hazard & exposure, vulnerability, and lack of coping capacity). We assume a linear mapping from the indicator to the level of pollution stock f, ranging from $f = 0$ (INFORM Risk indicator: 0, negligible pollution stock) to $f = 1$ (INFORM Risk indicator: 10, very high pollution stock). The level of pre-industrial pollution is approximated by the time average of the least polluted country in the sample (Cape Verde), setting $f_{pre} = 0.27$.

The Climate-driven INFORM Risk dataset provides eleven annual observations, subtracting f_{pre} we obtain $\{\widehat{f}(t_i)\}_{i=1}^{11}$. We assume that no country in the panel allocated a significant portion of its GDP to climate adaptation during the observation period. Therefore, the sample can be viewed as a realization of the process in (5) with $u = 0$.

Annual realizations $\widehat{f}(t_i)$ are normally distributed, with conditional expected value and unconditional variance:

$$\begin{aligned} \mathbb{E}[\widehat{f}(t_{i+1})|\widehat{f}(t_i)] &= e^{-\varphi}\widehat{f}(t_i) + k_e \int_{t_i}^{t_{i+1}} e^{-gs} y(s) e^{-\varphi(t_{i+1}-s)} ds \\ Var[\widehat{f}(t_i)] &= \frac{\sigma_f^2}{2\varphi}(1 - e^{-2\varphi}). \end{aligned} \tag{32}$$

The four parameters in (32) $(\varphi, \sigma_f, k_e, g)$ could be directly estimated using a Maximum Likelihood (ML) estimator on a country-by-country basis. However, to enhance the

[3]Source: `https://www.imf.org/external/datamapper/PPPPC@WEO/OEMDC/ADVEC/WEOWORLD`.
[4]Source: `https://climatedata.imf.org`.

SSA	LATAM	SA-APAC
Angola	Antigua and Barbuda	Bangladesh
Benin	Argentina	Bhutan
Botswana	Bahamas, The	Cambodia
Burkina Faso	Barbados	China
Burundi	Belize	Fiji
Cabo Verde	Bolivia	India
Cameroon	Brazil	Indonesia
Central African Republic	Chile	Kiribati
Comoros	Colombia	Lao PDR
Congo, Dem. Rep.	Costa Rica	Malaysia
Congo, Rep.	Dominica	Maldives
Cote d'Ivoire	Dominican Republic	Marshall Islands
Eswatini	Ecuador	Micronesia, Fed. Sts.
Ethiopia	El Salvador	Mongolia
Gabon	Grenada	Myanmar
Gambia, The	Guatemala	Nepal
Ghana	Guyana	Pakistan
Guinea	Haiti	Palau
Guinea-Bissau	Honduras	Papua New Guinea
Kenya	Jamaica	Philippines
Lesotho	Mexico	Samoa
Liberia	Nicaragua	Solomon Islands
Madagascar	Panama	Sri Lanka
Malawi	Paraguay	Thailand
Mali	Peru	Timor-Leste
Mauritania	St. Lucia	Tonga
Mauritius	Suriname	Tuvalu
Mozambique	Trinidad and Tobago	Vanuatu
Namibia	Uruguay	Viet Nam
Niger		
Nigeria		
Rwanda		
Senegal		
Seychelles		
Sierra Leone		
South Africa		
Sudan		
Tanzania		
Togo		
Uganda		
Zambia		
Zimbabwe		

Table A.1: List of developing and emerging countries split by macro-region